Cadmium intake and systemic exposure in postmenopausal women and age-matched men who smoke cigarettes.

Mean blood cadmium (B-Cd) concentrations are two- to threefold higher in smokers than in nonsmokers. The basis for this phenomenon is not well understood. We conducted a detailed, multifaceted study of cadmium exposure in smokers. Groups were older smokers (62±4 years, n = 25, 20% male) and nonsmokers (62±3 years, n = 16, 31% male). Each subject's cigarettes were machine smoked, generating individually paired measures of inhaled cadmium (I-Cd) versus B-Cd; I-Cd and B-Cd were each evaluated three times, at monthly intervals. Urine cadmium (U-Cd) was analyzed for comparison. In four smokers, a duplicate-diet study was conducted, along with a kinetic study of plasma cadmium versus B-Cd. Female smokers had a mean B-Cd of 1.21ng Cd/ml, with a nearly 10-fold range (0.29-2.74ng Cd/ml); nonsmokers had a lower mean B-Cd, 0.35ng Cd/ml ($p < 0.05$), and narrower range (0.20-0.61ng Cd/ml). Means and ranges for males were similar. Estimates of cadmium amounts inhaled daily for our subjects smoking ≥ 20 cigarettes/day were far less than the 15 µg Cd reported to be ingested daily via diet. This I-Cd amount was too low to alone explain the 3.5-fold elevation of B-Cd in our smokers, even assuming greater cadmium absorption via lungs than gastrointestinal tract; cadmium accumulated in smokers' lungs may provide the added cadmium. Finally, B-Cd appeared to be linearly related to I-Cd values in 75% of smokers, whereas 25% had far higher B-Cd, implying a possible heterogeneity among smokers regarding circulating cadmium concentrations and potentially cadmium toxicity.